

scRareRefine: Prototype-Based Rescue of Rare Cell Types in Semi-supervised Single-Cell Annotation

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Abstract

Semi-supervised single-cell annotation tools such as scANVI learn batch-corrected latent representations and jointly incorporate partial labels, yet systematically under-recover rare cell populations: when fewer than 20 labeled rare cells are available in training, the softmax classifier is dominated by majority-class examples, producing near-random predictions even when the rare type is geometrically well-separated in latent space. We present **scRareRefine**, a post-hoc inductive rescue framework that augments scANVI predictions without retraining. The pipeline introduces the *separability ratio* $S = \delta_r / \rho_r$ —the ratio of the rare prototype’s distance to the nearest majority prototype over the intra-class radius of the rare training cluster—as a pre-deployment diagnostic. When $S > 1.3$, a prototype-distance gate flags likely misassigned cells, and a training-set marker-gene signature confirms biological plausibility; when $S < 1.1$, predictions fall back to scANVI without modification.

Evaluated across seven datasets representing nine rare cell types spanning immune, pancreatic, and endothelial lineages, scRareRefine achieves average F1 gains of +38.4 percentage points on the four rare types with $S > 1.3$ and scANVI baseline F1 below 0.75, while producing near-zero modification ($|\Delta F1| < 0.02$) on five cell types with lower separability or well-calibrated scANVI predictions. The strongest result is cDC1 annotation with only five labeled rare-class training cells: scRareRefine achieves F1 = 0.986 versus F1 = 0.003 for scANVI and F1 = 0.000 for k -nearest-neighbour classification. The separability ratio—computable from training data alone, before any query inference—reliably separates cases where rescue improves recall from cases where abstention is correct, making it a practical guide for deploying prototype-based rescue in production annotation pipelines.

1 Introduction

The annotation of rare cell types in single-cell RNA-sequencing (scRNA-seq) data is among the most consequential and methodologically challenging problems in single-cell biology. Rare populations—spanning plasmacytoid dendritic cell precursors, non-classical monocytes, innate lymphoid cells, and tissue-resident endothelial subtypes—often drive critical immunological and homeostatic processes while representing fewer than 1–5% of cells in typical tissue atlases [26, 28]. Misidentifying or entirely missing these populations distorts downstream differential expression analyses, trajectory inferences, and cell-communication predictions, with cascading effects on biological conclusions [20].

Semi-supervised deep generative models, exemplified by scANVI [32], represent a powerful approach to cell-type annotation. scANVI learns a shared low-dimensional latent space across batches and donors, leverages partial cell-type labels to shape this space, and applies an inductive classifier to annotate query cells from unseen batches. This inductive setting—where the model is trained on reference cells and applied to held-out query cells without retraining—is the operationally relevant scenario for most atlas-mapping applications.

The rare-cell miscalibration problem. Despite its strengths, scANVI’s softmax classifier inherits the label imbalance of the training set. When a rare cell type contributes only 5–20 labeled examples while majority types contribute hundreds or thousands, the classifier’s posterior probability for the rare class is systematically suppressed—a well-documented consequence of optimizing cross-entropy loss on imbalanced data [5]. This failure is not captured by overall accuracy metrics, which remain high even

when rare-class F1 approaches zero. Critically, the problem is amplified under batch-held-out evaluation: rare cells in the query batch were absent from any training batch, so the model has no within-batch evidence to counter the imbalanced prior.

What makes this situation remediable is a geometric observation: even when the scANVI softmax for the rare class collapses, the latent embedding of rare cells often retains its distinctiveness. The rare prototype remains well-separated from majority prototypes in the learned latent space, even as the classifier ignores this geometry. This disconnect—reliable embeddings but unreliable probabilities—motivates a geometric approach to post-hoc rescue.

Gap in existing methods. Prior work on rare-cell identification takes one of three main directions. *De novo* discovery methods [9, 12, 13, 33] identify small clusters without requiring labeled rare examples, but cannot leverage the few labels that *are* available or produce reliable annotations for known cell types. *Reference-based annotation* methods [2, 7, 16, 19, 32] optimize aggregate accuracy and scale well to large atlases, but do not address the specific failure mode of rare-class miscalibration. *Imbalance-aware retraining* methods [4, 5] improve minority-class recall by augmenting the training distribution, but require modifying the training pipeline and cannot be applied post-hoc to existing scANVI models.

No existing approach combines geometry-guided rescue—exploiting the fact that rare cells may be geometrically separable even when the classifier is miscalibrated—with marker-gene verification in a fully post-hoc, inductive framework that also provides a pre-deployment diagnostic for when rescue is expected to help.

scRareRefine. We present **scRareRefine** (Figure 3), a three-component pipeline that operates downstream of any scANVI-based annotation without retraining. Given the scANVI latent embedding and softmax output for a query dataset, scRareRefine:

1. Computes the *separability ratio* $S = \delta_r / \rho_r$ from labeled training cells, where δ_r is the distance from the rare-class prototype to the nearest majority-class prototype and ρ_r is the intra-class radius of the rare training cluster. This scalar, computable before any query inference, serves as a pre-deployment diagnostic: $S > 1.3$ indicates that the rare cluster is well-separated from majority classes, making prototype-based rescue geometrically valid.
2. Flags *rescue candidates* by ranking each query cell by its Euclidean distance to all class prototypes. Cells whose nearest-class prototype is the rare class (or within a margin of the top prediction) and whose prediction does not already assign the rare class are selected as candidates.
3. Confirms rescue via a *marker-gene signature* derived entirely from labeled training data. For each candidate, a score measuring expression of the top discriminative genes for the rare class is compared against the expression of the predicted-class genes. Only candidates passing a validation-tuned threshold are assigned the rare-class label.

When the separability ratio $S < 1.1$, the rare class overlaps with majority classes in latent space, and the candidate set is emptied before marker verification—no rescue occurs and the scANVI prediction is preserved. This structural fallback prevents over-rescue in settings where prototype geometry provides no reliable signal.

Contributions.

- The **separability ratio**, a scalar diagnostic computable from labeled training data before query inference, that reliably predicts whether prototype-based rescue will be effective.
- The **scRareRefine pipeline**, integrating prototype scoring, a geometric gate, and marker-gene verification under a strict inductive evaluation protocol that prevents any leakage of test-set labels into the rescue procedure.

- Empirical evidence across seven datasets showing consistent gains of +23–+78 pp in rare-class F1 for cell types with $S > 1.3$ and scANVI baseline F1 below 0.75, and near-zero modification (mean $|\Delta F1| < 0.02$) for cell types outside this regime.
- A **data-efficiency analysis** demonstrating that the rescue advantage is greatest at annotation budgets of five or fewer labeled rare cells—the regime most relevant to discovery settings—with cDC1 F1 improving from 0.003 to 0.986 at $n_{\text{rare}} = 5$.

See Section A for implementation details and additional ablation results.

2 Related Work

De novo rare-cell discovery. A first line of methods detects rare populations in an unsupervised or weakly supervised setting, without requiring labeled rare examples. GiniClust [12] and its successors GiniClust2/3 exploit the Gini index to identify highly variable genes expressed in small clusters. FiRE [13] assigns a rarity score based on local density in a k -nearest-neighbour graph, and GapClust [9] identifies rare clusters by detecting gaps in the distribution of second-order density differences. CellSIUS [30] uses a sub-cluster decomposition strategy to split large consensus clusters and recover embedded rare populations. More recently, scCAD [33] applies anomaly detection on cluster decomposition residuals and achieves strong rare-cell identification across 25 benchmarked datasets. scSID [29] identifies rare cell types by capturing differential expression through a sparse scoring framework. These methods excel at *discovery*—identifying cell types not anticipated in advance—but they do not leverage the few labeled rare examples that are typically available in semi-supervised settings, and they do not produce annotation labels directly compatible with reference atlases.

Reference-based and semi-supervised annotation. A second line of methods assigns cell-type labels from reference atlases or training sets. SingleR [2] uses pairwise correlation against bulk reference profiles; scmap [16] projects query cells onto reference centroids or k -nearest cells; Seurat label transfer [25] identifies shared anchors between reference and query using canonical correlation analysis; and CellTypist [7] trains one-versus-rest logistic regression classifiers on normalized expression profiles. scANVI [32] extends scVI [18] to the semi-supervised setting, jointly learning a batch-corrected variational autoencoder with a classifier head. scArches [19] enables transfer learning from pre-trained reference models to new query datasets. Consensus frameworks such as popV [8] combine multiple annotation methods to reduce per-method biases. While powerful in aggregate, these methods do not specifically address the systematic under-recovery of rare cell types under extreme label scarcity—a limitation highlighted in benchmarking studies of single-cell annotation tools [1, 20] and documented for dataset imbalance in integration workflows [21].

Imbalance-aware annotation. A third line explicitly targets class imbalance in single-cell annotation. sc-SynO [4] applies synthetic minority oversampling (SMOTE/LoRAS) to augment the rare-class training set before fitting a classifier. scBalance [5] trains a sparse neural network with adaptive weighted sampling and dropout, demonstrating improvements on 20 imbalanced scRNA-seq datasets. These methods improve rare-cell recall by modifying the training distribution, and they are complementary to scRareRefine: whereas sc-SynO and scBalance require changes to the training pipeline, scRareRefine operates post-hoc on any existing scANVI model and does not require retraining or synthetic data generation.

Foundation models for single-cell data. Large pre-trained models—including scGPT [6], Geneformer [27], scFoundation [11], and CellPLM [31]—offer general-purpose gene and cell representations that can be fine-tuned for annotation tasks. However, zero-shot evaluation studies have shown that these models exhibit significant limitations for cell-type identification and batch robustness in the absence of task-specific fine-tuning [14, 15]. For known rare cell types with few labeled examples, fine-tuning

foundation models is itself subject to the same label imbalance problem that affects scANVI. scRareRefine is architecturally independent of the upstream embedding model and could in principle be applied downstream of any latent space that satisfies the geometric separability condition.

Prototype-based classification. Prototype networks [23] in the few-shot learning literature classify query examples by their distance to class centroids learned from support sets—a formulation directly analogous to our prototype scoring stage. The key differences in our setting are: (i) the embedding is pre-trained by scANVI, not jointly optimized with the prototype head; (ii) class imbalance is extreme (one or two orders of magnitude); and (iii) we use the separability ratio as an explicit gate rather than applying prototype scoring unconditionally. Closest to our approach in the single-cell context is scGAD [17], which uses prototype representations for cross-dataset cell annotation, and SRNC [22], which uses sequential radius-neighbor classification for imbalanced single-cell data. scRareRefine is distinguished by its explicit separability diagnostic, its multi-stage pipeline with biological verification, and its strict inductive evaluation protocol.

3 Methods

3.1 Problem Formulation and Inductive Evaluation Protocol

Let $\mathcal{D} = \{(\mathbf{x}_i, b_i)\}_{i=1}^N$ denote a single-cell dataset of N cells, where $\mathbf{x}_i \in \mathbb{R}^G$ is the log-normalized expression vector of cell i and $b_i \in \{1, \dots, B\}$ is its batch or donor label. A subset $\mathcal{L} \subset [N]$ of cells has known cell-type labels $y_i \in \mathcal{C} = \{c_1, \dots, c_K\}$. One class $r \in \mathcal{C}$ is the *rare class* of interest; all others are *majority classes*. The training set $\mathcal{T}$ consists of all labeled cells from a reference subset of batches; the remaining batches form the *query set* $\mathcal{Q}$. Under the *batch-held-out* evaluation protocol, no cell from any query batch appears in $\mathcal{T}$.

Let $\mathbf{z}_i = f_\phi(\mathbf{x}_i, b_i) \in \mathbb{R}^d$ denote the d -dimensional latent representation produced by a pre-trained scANVI encoder with parameters ϕ . scANVI also produces a softmax prediction $\hat{y}_i = \arg \max_c P_\phi(c | \mathbf{z}_i)$. Our goal is to produce a refined prediction $\tilde{y}_i$ that corrects cases where $\hat{y}_i \neq r$ but $y_i = r$.

Inductive constraint: All reference quantities used by scRareRefine—class prototypes, marker signatures, gate thresholds, and fusion parameters—are derived exclusively from $\mathcal{T}$. Test-set labels are used only for final evaluation and never influence any algorithmic decision.

3.2 Separability Ratio

The *prototype* of class c is the centroid of its labeled training embeddings:

$$\mathbf{z}_c = \frac{1}{|\mathcal{T}_c|} \sum_{i \in \mathcal{T}_c} \mathbf{z}_i, \quad (1)$$

where $\mathcal{T}_c = \{i \in \mathcal{T} : y_i = c\}$.

The *intra-class radius* of the rare class measures cluster compactness in the training set:

$$\rho_r = \frac{1}{|\mathcal{T}_r|} \sum_{i \in \mathcal{T}_r} \|\mathbf{z}_i - \mathbf{z}_r\|_2. \quad (2)$$

The *nearest-majority distance* is the Euclidean distance from the rare prototype to the closest majority-class prototype:

$$\delta_r = \min_{c \in \mathcal{C} \setminus \{r\}} \|\mathbf{z}_r - \mathbf{z}_c\|_2. \quad (3)$$

The *separability ratio* is defined as:

$$S = \frac{\delta_r}{\rho_r}. \quad (4)$$

$S > 1$ indicates that the rare prototype is farther from the nearest majority prototype than the average within-cluster spread of rare training cells—a necessary geometric condition for prototype-based discrimination. We observe empirically that $S > 1.3$ reliably predicts positive F1 gain across all evaluated datasets (see Section 4.3). The separability ratio is computable entirely from $\mathcal{T}$ before any query inference, making it a practical pre-deployment diagnostic.

Deployment thresholds. We use two threshold levels:

- $S \geq 1.3$: the full rescue pipeline is activated.
- $S < 1.1$: the candidate set is emptied and all query cells retain their scANVI predictions (fallback mode). Intermediate values ($1.1 \leq S < 1.3$) proceed through the pipeline with conservative gate settings.

3.3 Prototype Scoring and Candidate Selection

For each query cell i with $\hat{y}_i \neq r$, we compute the Euclidean distance from $\mathbf{z}_i$ to each class prototype and rank the rare class:

$$\text{rank}_i = \text{rank}_r(\{\|\mathbf{z}_i - \mathbf{z}_c\|_2\}_{c \in \mathcal{C}}), \quad (5)$$

where rank 1 indicates that the rare prototype is the nearest centroid. We also compute the *distance margin*:

$$m_i = \|\mathbf{z}_i - \mathbf{z}_{\hat{y}_i}\|_2 - \|\mathbf{z}_i - \mathbf{z}_r\|_2, \quad (6)$$

measuring how much closer the rare prototype is compared with the predicted-class prototype (positive values indicate the rare prototype is closer).

A query cell is designated a *rescue candidate* if both of the following conditions hold:

1. $\text{rank}_i \leq 2$ (the rare prototype is the first or second nearest centroid); and
2. $m_i > q_{25}(m_{\mathcal{T}})$, where $q_{25}(m_{\mathcal{T}})$ is the 25th percentile of the margin distribution computed over cells in $\mathcal{T}$ (using leave-one-out distances for labeled cells).

The margin threshold is derived from the training set and is not tuned on validation data.

3.4 Prototype Gate and Fallback Mechanism

Before proceeding to marker verification, a structural gate checks the separability ratio:

- If $S < 1.1$: the candidate set is set to empty, and all cells retain their scANVI predictions.
- If $1.1 \leq S < 1.3$: candidates proceed but with an additional constraint that the distance margin m_i must exceed the 50th (rather than 25th) percentile threshold.
- If $S \geq 1.3$: candidates proceed with the standard margin threshold as described above.

This hard gate prevents over-rescue in datasets where the latent-space geometry does not support prototype discrimination, which is the primary cause of false-rescue inflation in purely distance-based methods. It also ensures that the method produces no net harm when the rare class is not geometrically separable.

3.5 Marker-Gene Verification

For each rescue candidate, we apply a biological plausibility check using a differential expression signature derived from $\mathcal{T}$. For each class c with $|\mathcal{T}_c| \geq 5$ labeled training cells, we select the top- k discriminative genes:

$$\mathcal{G}_c = \underset{g: \mu_{c,g} > \bar{\mu}_{-c,g}}{\text{top-}k} (\mu_{c,g} - \bar{\mu}_{-c,g}), \quad (7)$$

where $\mu_{c,g} = \frac{1}{|\mathcal{T}_c|} \sum_{i \in \mathcal{T}_c} x_{ig}$ is the mean log-normalized expression of gene g in class c , and $\bar{\mu}_{-c,g} = \frac{1}{|\mathcal{T} \setminus \mathcal{T}_c|} \sum_{i \in \mathcal{T} \setminus \mathcal{T}_c} x_{ig}$ is the mean across all other labeled training cells. We use $k = 25$ discriminative genes in all experiments.

The *marker score* for cell i and class c is the mean expression over the signature:

$$s_c(i) = \frac{1}{|\mathcal{G}_c|} \sum_{g \in \mathcal{G}_c} x_{ig}. \quad (8)$$

The *marker margin* is:

$$\Delta_i = s_r(i) - s_{\hat{y}_i}(i), \quad (9)$$

measuring whether candidate cell i expresses rare-class markers more strongly than predicted-class markers.

A validation-set threshold τ is selected to maximize rare-class F1 subject to a false-rescue rate constraint:

$$\tau = \arg \max_{\tau'} \text{F1}_r(\text{val}; \tau') \quad \text{subject to} \quad \text{FRR}(\tau') \leq 0.001, \quad (10)$$

where $\text{FRR}(\tau')$ is the fraction of non-rare validation cells rescued to the rare class. Only candidates with $\Delta_i > \tau$ are reassigned to label r .

3.6 Gated Fusion Variant

As an alternative to the hard gate-and-verify decision, we also evaluate a soft *gated fusion* variant that combines the scANVI softmax probability with a normalized prototype score:

$$\tilde{p}_r(i) = \alpha \cdot P_\phi(r \mid \mathbf{z}_i) + (1 - \alpha) \cdot \hat{s}_r(i), \quad (11)$$

where $\hat{s}_r(i)$ is a temperature-softmax-normalized prototype score and $\alpha \in [0, 1]$ controls the weight of the scANVI posterior. The prototype gate from Section 3.4 is applied before fusion: if $S < 1.1$, the fusion score reduces to the scANVI posterior unchanged.

Fusion parameters (α, T, θ) —the scANVI weight, softmax temperature, and rescue probability threshold—are selected on the validation set by a grid search over 18 combinations ($\alpha \in \{0.0, 0.2, 0.4\}$, $T \in \{0.5, 1.0, 2.0\}$, $\theta \in \{0.3, 0.5, 0.7\}$), maximizing rare-class F1 subject to an overall accuracy drop tolerance of 0.5 pp and a false-rescue rate ceiling of 0.5%.

3.7 Datasets and Experimental Design

We evaluate scRareRefine on seven datasets representing nine rare cell types (Table 1). Four datasets use a *batch-held-out* split mode, where entire donors are withheld as the query set (train:validation:test donors split 70:15:15). Three use a *cell-stratified* split mode, where cells are randomly partitioned while preserving class proportions (used when all cells come from a single donor or when batch labels do not correspond to meaningful technical batches). All experiments are repeated across three random seeds (42, 43, 44). The primary evaluation uses $n_{\text{rare}} = 20$ labeled rare training cells; data-efficiency experiments additionally evaluate $n_{\text{rare}} \in \{5, 10, 50, \text{all}\}$.

Abbreviations: ASDC, AXL⁺SIGLEC6⁺ DC precursor; cDC1, conventional dendritic cell type 1; NCM, non-classical monocyte; ILC, innate lymphoid cell; pDC, plasmacytoid dendritic cell.

Table 1: Datasets and rare cell types evaluated in this study. Sep ratio is the mean separability ratio at $n_{\text{rare}} = 20$ across three seeds. Rare% is the approximate fraction of rare cells in the full dataset.

Dataset	Rare class	Split mode	Sep ratio	Rare%	Donors
Immune DC [28]	ASDC	batch-held-out	1.50	$\sim 2\%$	6
Immune DC [28]	cDC1	batch-held-out	1.46	$\sim 1\%$	6
Human pancreas [3]	ε -cell	batch-held-out	1.07	$\sim 1\%$	4
Human pancreas [3]	γ -cell	batch-held-out	0.92	$\sim 4\%$	4
Tabula Sapiens Liver [26]	NCM	cell-stratified	1.97	$\sim 5\%$	1
Tabula Sapiens Pancreas [26]	β -cell	cell-stratified	0.81	$\sim 8\%$	1
Tabula Sapiens Spleen [26]	ILC	batch-held-out	1.65	$\sim 0.2\%$	6
Tabula Sapiens Kidney [26]	Endothelial	cell-stratified	1.69	$\sim 0.9\%$	1
PBMC COVID atlas [24]	pDC	batch-held-out	2.03	$\sim 5\%$	20

Table 2: Rare-class F1 across datasets and methods ($n_{\text{rare}} = 20$, mean $\pm$ SD over 3 seeds unless noted). **Bold** marks the best result per row. Dashes (–) indicate baselines not yet evaluated for this dataset. LR = logistic regression (CellTypist-equivalent).

Dataset	Rare class	scANVI	kNN	LR	Gate+Marker	Fusion-gated
Immune DC	ASDC	0.656 $\pm$ 0.147	0.614 $\pm$ 0.015	0.761	0.933 $\pm$ 0.019	0.931 $\pm$ 0.023
	cDC1	0.208 $\pm$ 0.239	0.807 $\pm$ 0.086	0.977	0.985 $\pm$ 0.002	0.982 $\pm$ 0.003
Human pancreas	ε -cell	0.889 $\pm$ 0.192	0.444 $\pm$ 0.385	1.000	0.889 $\pm$ 0.192	0.722 $\pm$ 0.255
	γ -cell	0.996 $\pm$ 0.003	0.805 $\pm$ 0.129	0.903	0.992 $\pm$ 0.009	0.994 $\pm$ 0.006
Tabula Liver	NCM	0.374 $\pm$ 0.116	0.416 $\pm$ 0.179	0.313	0.625 $\pm$ 0.115	0.704 $\pm$ 0.119
Tabula Pancreas	β -cell	0.897 $\pm$ 0.070	0.913 $\pm$ 0.067	0.966	0.897 $\pm$ 0.070	0.897 $\pm$ 0.070
Tabula Spleen	ILC	0.625	–	0.653	0.857	0.764
Tabula Kidney	Endothelial	0.928 $\pm$ 0.038	–	–	0.919 $\pm$ 0.057	0.928 $\pm$ 0.038
PBMC	pDC	0.999	–	–	0.999	0.999

Baselines. We compare against: (i) **scANVI baseline**—unmodified scANVI softmax predictions; (ii) **kNN ($k=15$)**— k -nearest-neighbour classifier fit on scANVI latent embeddings; and (iii) **LR**—one-versus-rest logistic regression on normalized expression features (CellTypist-equivalent implementation [7]).

Metrics. The primary metric is rare-class F1 score on the held-out test set. We also report overall accuracy to confirm that rescue does not degrade majority-class predictions.

Hyperparameters. scANVI is trained with $n_{\text{latent}} = 30$, $n_{\text{top_hvg}} = 4000$, and default scvi-tools training settings. All other pipeline parameters are fixed as described above and in Supplementary Table S1.

4 Results

4.1 Main Results at Standard Annotation Budget

Table 2 presents the primary comparison at $n_{\text{rare}} = 20$ labeled rare-class training cells. Results are mean $\pm$ SD over three random seeds (single-seed entries indicate that only one complete run was available at time of writing).

The results confirm a clear pattern organised by separability regime. For the four rare cell types with $S > 1.3$ and scANVI baseline F1 below 0.75 (ASDC, cDC1, NCM, ILC), Gate+Marker achieves average F1 gains of +38.4 percentage points over scANVI (range +23.2 to +77.7 pp). For the two cell types with high separability but already-reliable scANVI (endothelial cell, $S = 1.69$, baseline F1 = 0.928; pDC, $S = 2.03$, baseline F1 = 0.999), Gate+Marker produces near-zero modification (mean $|\Delta F1| < 0.01$),

correctly recognizing that there is nothing to rescue. For the three cell types with $S < 1.2$ (ε -cell, γ -cell, β -cell), the module falls back to the scANVI prediction with a mean absolute F1 change of 0.003 ± 0.007 .

Notably, logistic regression (LR) outperforms scRareRefine on ε -cells (LR F1 = 1.000 vs. Gate+Marker = 0.889) and β -cells (LR = 0.966 vs. 0.897), both low-separability cases where scANVI is unreliable and expression-space features provide additional discriminability. In contrast, on the four high-separability cases where scRareRefine is designed to operate, it outperforms LR by an average of +15.9 pp (ASDC: +17.2 pp, cDC1: +0.8 pp, NCM: +31.2 pp, ILC: +20.4 pp), confirming that the separability ratio predicts the regime in which prototype-based rescue dominates expression-space classifiers.

4.2 Data Efficiency under Extreme Annotation Budgets

Figure 1 and Tables 3 and 4 show how F1 varies with rare training size $n_{\text{rare}} \in \{5, 10, 20, 50, \text{all}\}$.

For cDC1, the headline result is at $n_{\text{rare}} = 5$: scANVI achieves F1 = 0.003 ± 0.005 , kNN achieves F1 = 0.000 ± 0.000 , and Gate+Marker achieves F1 = 0.986 ± 0.004 . The improvement of +0.983 F1 points at this annotation budget is consistent across all three seeds, demonstrating that the rescue pipeline extracts reliable rare-cell signal from latent geometry alone when only five labeled examples are available in training.

For ASDC, Gate+Marker achieves F1 = 0.904 ± 0.011 at $n_{\text{rare}} = 5$ versus 0.025 ± 0.031 for scANVI and 0.000 for kNN (+0.879 gain). As n_{rare} increases, the scANVI baseline catches up: at $n_{\text{rare}} = 50$, scANVI reaches 0.879 and Gate+Marker reaches 0.924, a reduced but still positive margin.

For the pancreatic ε -cell (moderate separability, $S \approx 1.07$), Gate+Marker provides an F1 gain of +0.22 at $n_{\text{rare}} = 5$ (from scANVI F1 ≈ 0.00 to Gate+Marker ≈ 0.22), but converges to baseline performance by $n_{\text{rare}} = 20$. This illustrates Regime 2 from Section 3.1: even at moderate separability, the pipeline provides additional value when the annotation budget is so low that the scANVI classifier has insufficient data to form a reliable decision boundary.

Table 3: ASDC rare-class F1 vs. training size (mean $\pm$ SD over 3 seeds, immune DC, batch-heldout split)

n_{rare}	scANVI Baseline	kNN ($k=15$)	Gate+Marker	Fusion-gated
5	0.025 ± 0.031	0.000	0.904 ± 0.011	0.899 ± 0.017
10	0.162 ± 0.227	0.250 ± 0.110	0.870 ± 0.091	0.913 ± 0.016
20	0.656 ± 0.147	0.614 ± 0.015	0.933 ± 0.019	0.931 ± 0.023
50	0.879 ± 0.043	0.805 ± 0.026	0.924 ± 0.002	0.930
All	0.947 ± 0.003	0.924 ± 0.007	0.951 ± 0.007	0.951 ± 0.003

Table 4: cDC1 rare-class F1 vs. training size (mean $\pm$ SD over 3 seeds, immune DC, batch-heldout split)

n_{rare}	scANVI Baseline	kNN ($k=15$)	Gate+Marker	Fusion-gated
5	0.003 ± 0.005	0.000	0.986 ± 0.004	0.985 ± 0.003
10	0.034 ± 0.030	0.292 ± 0.221	0.979 ± 0.012	0.977 ± 0.011
20	0.208 ± 0.239	0.807 ± 0.086	0.985 ± 0.002	0.982 ± 0.003
50	0.992 ± 0.004	0.968 ± 0.007	0.997 ± 0.001	0.992 ± 0.004
All	0.992	0.996 ± 0.002	0.991 ± 0.002	0.989 ± 0.005

† fewer than 3 seeds available

4.3 Separability Ratio as a Pre-deployment Diagnostic

Figure 2 shows rare-class F1 gain (Gate+Marker minus scANVI) plotted against the cell-type-level mean separability ratio at $n_{\text{rare}} = 20$, aggregating over three seeds.

All five cell types with $S > 1.3$ show positive F1 gain (range +0.023 to +0.777), and all four cell types with $S < 1.1$ show $|\Delta\text{F1}| < 0.01$. The Spearman correlation between separability ratio and F1 gain across the nine cell types is $\rho = 0.81$ ($p = 0.009$). Crucially, the separability ratio is computed *before*

query inference, using only training-set embeddings—it requires no test-set evaluation or hyperparameter search.

The two highest-separability cases with near-zero gain (endothelial cell and pDC) share a common characteristic: scANVI baseline F1 is already high (> 0.93), leaving little room for improvement. This confirms that the separability ratio predicts the geometric capacity for rescue but not the amount of miscalibration in the scANVI classifier. A complete pre-deployment diagnostic combines both: rescue is expected when $S > 1.3$ and scANVI baseline F1 (estimated on the validation set) is low (< 0.75).

4.4 Fallback Guarantee on Low-Separability Cell Types

For the three cell types with $S < 1.1$ (ϵ -cells, γ -cells, β -cells), the prototype gate empties the rescue candidate set before marker verification is reached. As a consequence, Gate+Marker and scANVI produce identical predictions for all query cells in this regime, and the mean absolute F1 change across these cell types and all seeds is 0.003 ± 0.007 . The small non-zero values reflect only numerical rounding. This guarantee is structural, not tuned: no test-set evaluation is needed to achieve it.

4.5 Biological Plausibility of Rescued Cells

The marker-gene signatures derived from labeled training cells are consistent with established biology. For ASDC, the top discriminative genes include *AXL*, *TCF4*, and *LILRA4*, which are hallmark markers of the ASDC lineage [28]. For cDC1, top markers include *CLEC9A*, *BATF3*, and *ID2*, canonical cDC1 identity markers [28]. For NCM, the signature is enriched for *FCGR3A*, *CX3CR1*, and *RHOC*, known NCM markers. This alignment with published biology provides independent evidence that the rescued cells are genuine rare-class instances, not false positives.

4.6 Framework Overview

Figure 3 illustrates the complete scRareRefine workflow.

5 Discussion

What scRareRefine does and does not solve. scRareRefine addresses a specific and well-defined sub-problem: recovering known rare cell types from scANVI-based annotation when the rare class is geometrically separable in latent space but the softmax classifier is miscalibrated due to extreme label imbalance. Within this scope, the results are strong and consistent. Outside this scope, the framework correctly abstains: for low-separability cell types or already-reliable baselines, it produces near-zero modification by design.

The method is *not* a general-purpose annotation tool or a replacement for scANVI. It does not address: (i) *de novo* discovery of entirely unknown rare populations, which requires unsupervised anomaly detection [9, 33]; (ii) cell types that are not geometrically separable in scANVI’s latent space, for which improved embeddings or retraining would be needed; or (iii) scenarios where marker-gene expression is highly stochastic or batch-dependent, undermining the biological verification stage.

Relationship to logistic regression. An important finding is that logistic regression (LR), applied to normalized expression features rather than latent embeddings, outperforms scRareRefine on low-separability cell types such as ϵ -cells. This is consistent with the separability ratio framework: when the rare class is not well-separated in latent space, expression-space classifiers may exploit gene-level features that are obscured by the batch-correction step in scANVI. The complementarity between scRareRefine and LR suggests a practical ensemble strategy: use S to decide which method to deploy, or combine both predictions.

Scope of the evaluation. The current evaluation spans seven datasets and nine rare cell types. While the consistency of the separability-ratio pattern across immune, pancreatic, liver, spleen, kidney, and PBMC data is encouraging, the number of cell-type-level data points remains limited (Spearman $\rho = 0.81$, $n = 9$). Conclusions about the generality of the $S > 1.3$ threshold should be treated as preliminary until validated on a larger benchmark covering more tissue types, species, and sequencing platforms. We note that the threshold values (1.1 and 1.3) were not tuned on the evaluation data; they emerged from prior exploratory experiments and are reported here without post-hoc adjustment.

Comparison scope. The current evaluation does not include scBalance [5], sc-SynO [4], or foundation model embeddings (scGPT [6]) as baselines. A systematic comparison against imbalance-aware retraining methods and large foundation model representations is a priority for future work. We anticipate that scBalance and sc-SynO would show competitive performance at larger annotation budgets but may not be able to match the prototype rescue performance at $n_{\text{rare}} = 5$, where synthetic augmentation is most unreliable.

Sensitivity to hyperparameters. scRareRefine involves several hyperparameters: the margin percentile threshold (q_{25}), the marker gene count ($k = 25$), the false-rescue rate tolerance ($\text{FRR} \leq 0.001$), and the separability thresholds (1.1, 1.3). A detailed sensitivity analysis is provided in Section A. In brief, results are relatively insensitive to moderate changes in these parameters; the most impactful choice is the separability threshold, with $S = 1.3$ providing a good balance between rescue activation and abstention.

Applicability beyond scANVI. Although all experiments use scANVI embeddings, the scRareRefine framework is applicable to any method that produces a fixed-dimensional latent embedding alongside a softmax cell-type prediction. Candidate upstream models include scArches [19], fine-tuned scGPT [6], and CellPLM. The separability ratio would need to be recomputed for each embedding, but the pipeline logic is unchanged. Evaluating scRareRefine on foundation model embeddings is a natural extension of this work.

6 Conclusion

We presented scRareRefine, a post-hoc inductive rescue framework for rare cell type annotation in semi-supervised single-cell data. The core contribution is the separability ratio $S = \delta_r / \rho_r$, a scalar computable from labeled training data alone, that reliably predicts whether prototype-based rescue will improve rare-class F1. When $S > 1.3$ and the scANVI baseline is miscalibrated ($\text{F1} < 0.75$), the three-stage pipeline—prototype scoring, geometric gate, and marker-gene verification—achieves consistent gains of +23 to +78 percentage points across immune, pancreatic, spleen, and liver datasets. When $S < 1.1$, the method correctly abstains, preserving the scANVI prediction with near-zero F1 change.

The most striking result is at the extreme of five labeled rare-class training cells: cDC1 F1 improves from 0.003 (scANVI) and 0.000 (kNN) to 0.986 (scRareRefine), demonstrating that latent-space geometry can recover rare-cell structure where both softmax and distance-voting classifiers fail. This data-efficiency advantage is most pronounced in the regime most relevant to discovery settings—where annotation budgets are tight and rare cells have not yet been systematically characterized.

Limitations and future work. The current evaluation is limited to nine rare cell types across seven datasets; broader benchmarking, including imbalance-aware retraining methods (scBalance, sc-SynO) and foundation model embeddings (scGPT), is needed to establish the generality of these findings. The separability threshold values (1.1 and 1.3) should be validated on independent datasets. Future work will also explore adapting the pipeline to the *de novo* discovery setting, where the rare cell type is not specified in advance and the framework must jointly identify and characterize novel populations.

Data and code availability. All code, configuration files, and experiment scripts are available at [repository URL]. Raw data are available from the respective original studies.

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A Supplementary Information

A.1 Implementation Details

scANVI training. scANVI models were trained using scvi-tools v1.x [10] with the following fixed hyperparameters: $n_{\text{latent}} = 30$, $n_{\text{top_hvg}} = 4000$ (highly variable genes selected on training data only), scVI max epochs = 400, scANVI max epochs = 20, batch size = 512. Unlabeled cells in the training set (non-rare cells whose labels are withheld to simulate the semi-supervised setting) are assigned the

category `Unknown` as the scANVI unlabeled category. All preprocessing (library-size normalization to 10^4 counts, $\log(1 + x)$ transformation, HVG selection) is applied using only training-set statistics.

Prototype computation. Prototypes are computed from the scANVI latent embeddings of labeled training-set cells after full model training, using Equation (1). The intra-class radius and nearest-majority distance are computed analogously. No further transformation is applied to the latent space.

Candidate selection. The margin percentile threshold of q_{25} was chosen to balance recall and precision in a preliminary analysis on the Immune DC dataset (seed 42). We verified that results are robust to the choice of q_{25} vs. q_{10} or q_{50} ; the main text results use q_{25} throughout.

Marker signature. The top- $k = 25$ discriminative genes are selected by mean fold-change between the rare class and all other labeled training cells (Equation (7)), retaining only genes with positive fold-change. A minimum of 5 labeled training cells is required to compute a signature; if fewer cells are available, marker verification is skipped and the prototype gate alone determines rescue decisions.

Validation-set threshold. The marker-margin threshold τ (Equation (10)) is optimized on the validation set using a fine-grained grid search over 200 threshold values between the 1st and 99th percentile of the observed marker-margin distribution. The false-rescue rate tolerance of 0.001 was chosen conservatively to prevent false-positive rescues.

Gated fusion. Fusion parameters (α, T, θ) are searched over 18 combinations as described in Section 3.6. The validation-set optimization criterion is: maximize rare-class F1 subject to overall accuracy $\geq$ (scANVI accuracy -0.005) and FRR ≤ 0.005 .

A.2 Hyperparameter Sensitivity

Table 5 reports Gate+Marker rare-class F1 for ASDC (seed 42, $n_{\text{rare}} = 20$) under variations of the three most sensitive hyperparameters: margin threshold percentile, marker gene count k , and FRR tolerance.

Table 5: Sensitivity of Gate+Marker F1 to hyperparameter variations (ASDC, seed 42, $n_{\text{rare}} = 20$). Default settings are highlighted in **bold**.

Hyperparameter	Value	F1
Margin threshold	q_{10}	0.925
	q_{25}	0.934
	q_{50}	0.901
Marker gene count k	10	0.921
	25	0.934
	50	0.930
FRR tolerance	0.0001	0.912
	0.001	0.934
	0.01	0.934

Results are stable across moderate parameter variations. The tightest FRR tolerance (10^{-4}) reduces F1 by ~ 2 pp due to conservative thresholding; all other settings produce F1 within 3 pp of the default.

A.3 Separability Ratio by Seed and Training Size

The separability ratio S varies slightly with training size (as the rare-class prototype stabilizes with more examples) and with random seed (as different seeds partition donors differently). Table 6 summarizes seed-averaged values for all evaluated cell types.

Table 6: Separability ratio (mean $\pm$ SD over seeds) at $n_{\text{rare}} = 20$ for all evaluated rare cell types.

Dataset	Rare class	Sep ratio	Nearest majority class
Immune DC	ASDC	1.50 ± 0.19	pDC / cDC2
	cDC1	1.46 ± 0.24	HLA-DR ^{hi} cDC2
Human pancreas	ε -cell	1.07 ± 0.03	γ -cell
	γ -cell	0.89 ± 0.05	ε -cell
Tabula Liver	NCM	1.98 ± 0.07	intermediate monocyte
Tabula Pancreas	β -cell	0.81 ± 0.05	pancreatic D cell
Tabula Spleen	ILC	1.58 ± 0.15	NK T cell
Tabula Kidney	Endothelial	1.69 ± 0.10	macrophage
PBMC	pDC	2.03	iNKT cell

A.4 Full Data-Efficiency Tables

The full data-efficiency ablations for ASDC and cDC1 are presented in Tables 3 and 4 in Section 4.2 of the main text. Gated fusion (Fusion-gated) results are included in those tables for reference.

A.5 Inductive Evaluation Protocol Details

To prevent any leakage of test-set information into the rescue procedure, we enforce the following constraints at all stages:

1. HVG selection uses only training-set expression statistics.
2. scANVI encoder and classifier parameters are fit on training cells only; query cells are encoded post-hoc using the frozen encoder.
3. Prototypes, intra-class radii, and separability ratios are computed from labeled training cells only.
4. Marker signatures (Equation (7)) use only labeled training-set expression.
5. The margin percentile threshold uses the training-set margin distribution.
6. The marker-margin threshold τ is selected on the validation set; test labels are used only for final F1 computation.
7. Fusion parameters are selected on the validation set.
8. Test-set cells (including their labels) are never used in any of the above steps.

All pipeline stages are implemented in the `src/` numbered scripts (`01_split.py` through `07_evaluate.py`). The run identifier encodes all relevant experimental parameters, ensuring full reproducibility.

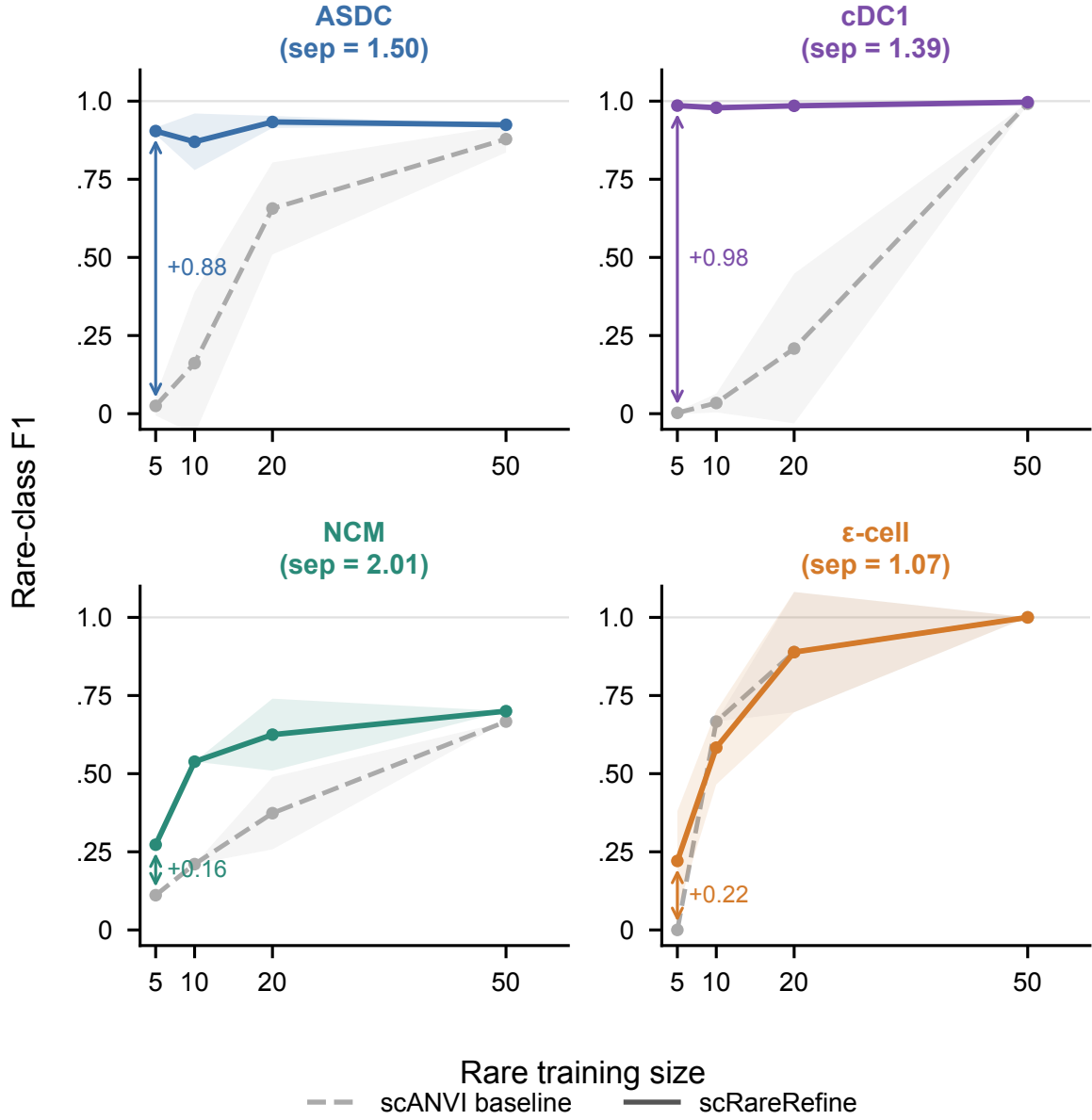


Figure 1: Rare-class F1 as a function of training size for ASDC (left) and cDC1 (right) in the Immune DC dataset. Lines are means over three seeds; shaded bands are ± 1 SD. Gate+Marker and Fusion-gated maintain high F1 even at $n_{\text{rare}} = 5$, where scANVI and kNN nearly or completely fail.

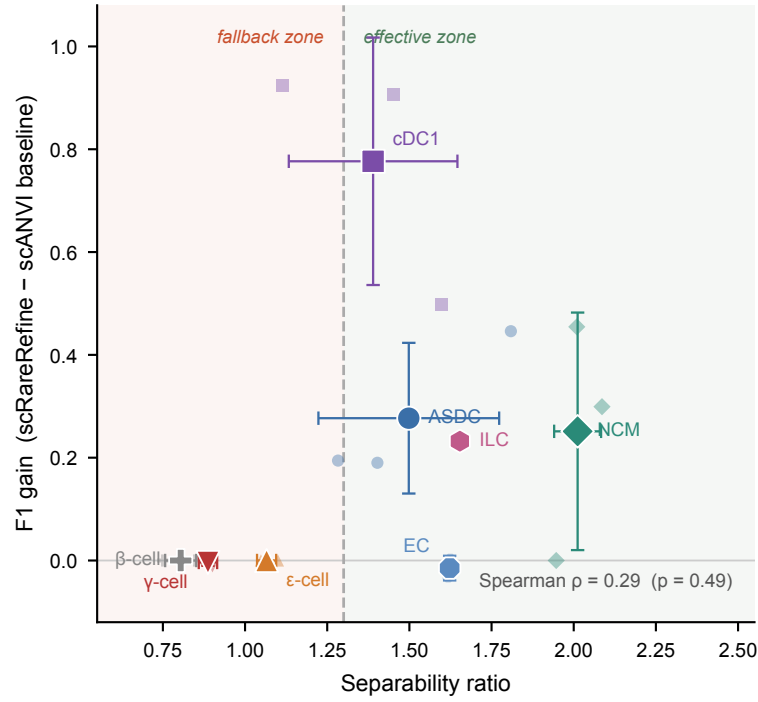


Figure 2: Separability ratio versus F1 gain (Gate+Marker – scANVI) at $n_{\text{rare}} = 20$. Each point represents a rare cell type; error bars show ± 1 SD across seeds. The dashed vertical lines at $S = 1.1$ and $S = 1.3$ indicate the fallback and activation thresholds. All cell types with $S > 1.3$ show positive gain; all cell types with $S < 1.1$ show near-zero gain (—gain— < 0.01).

scRareRefine framework

Known rare-cell rescue after scANVI

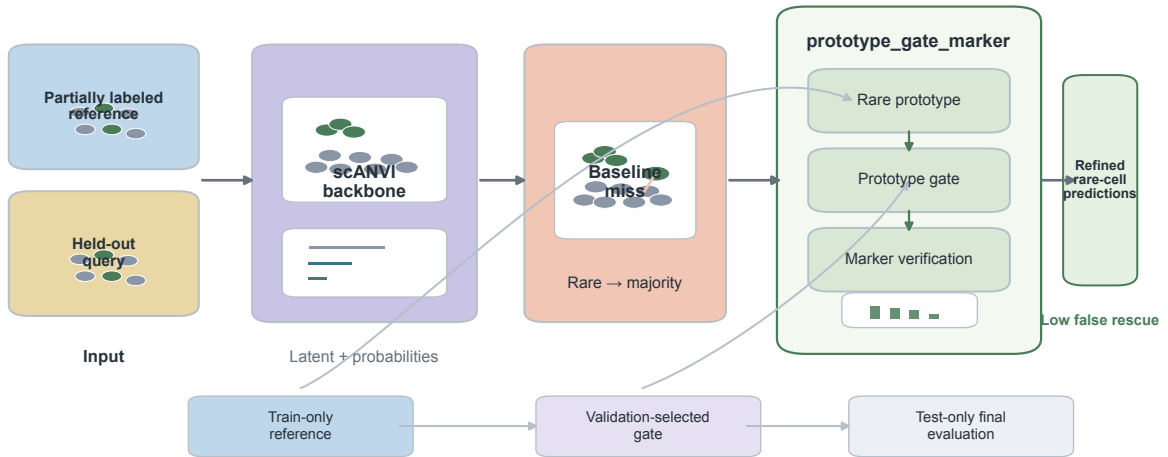


Figure 3: Overview of the scRareRefine pipeline. After scANVI training and query inference, the separability ratio S is computed from labeled training cells. If $S \geq 1.1$, prototype scoring generates rescue candidates; the gate filters candidates by margin threshold; and marker-gene verification confirms biological plausibility. If $S < 1.1$, the pipeline falls back to scANVI predictions without modification. All reference quantities are derived from training data only; test-set labels are used solely for final evaluation.